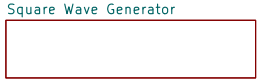
Power



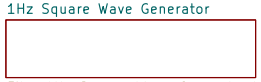
File: Power.kicad_sch

Square Wave Generator



File: Square_Wave_Generator.kicad_sch

1Hz Square Wave Generator



File: 1Hz_Square_Wave_Generator.kicad_sch

Seconds Counter



File: Seconds_Counter.kicad_sch

Minutes Counter



File: Minutes_Counter.kicad_sch

Hours Counter



File: Hours_Counter.kicad_sch

Colon LEDs



File: Colon LEDs.kicad_sch

Time Set Buttons



File: Time_Set_Buttons.kicad_sch

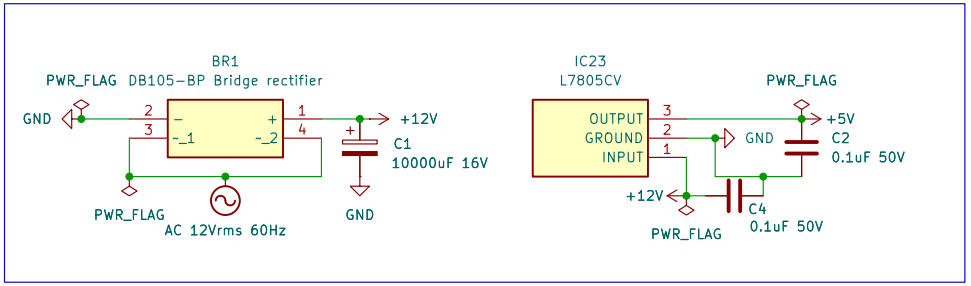
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /
File: Breadboard_Clock_24h.kicad_sch

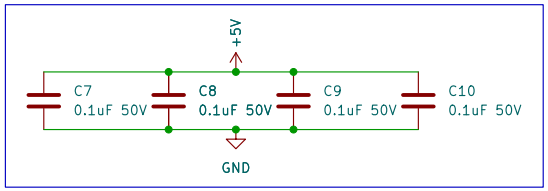
Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/9

12V DC and 5V DC



Extra Power Rail Noise Reduction



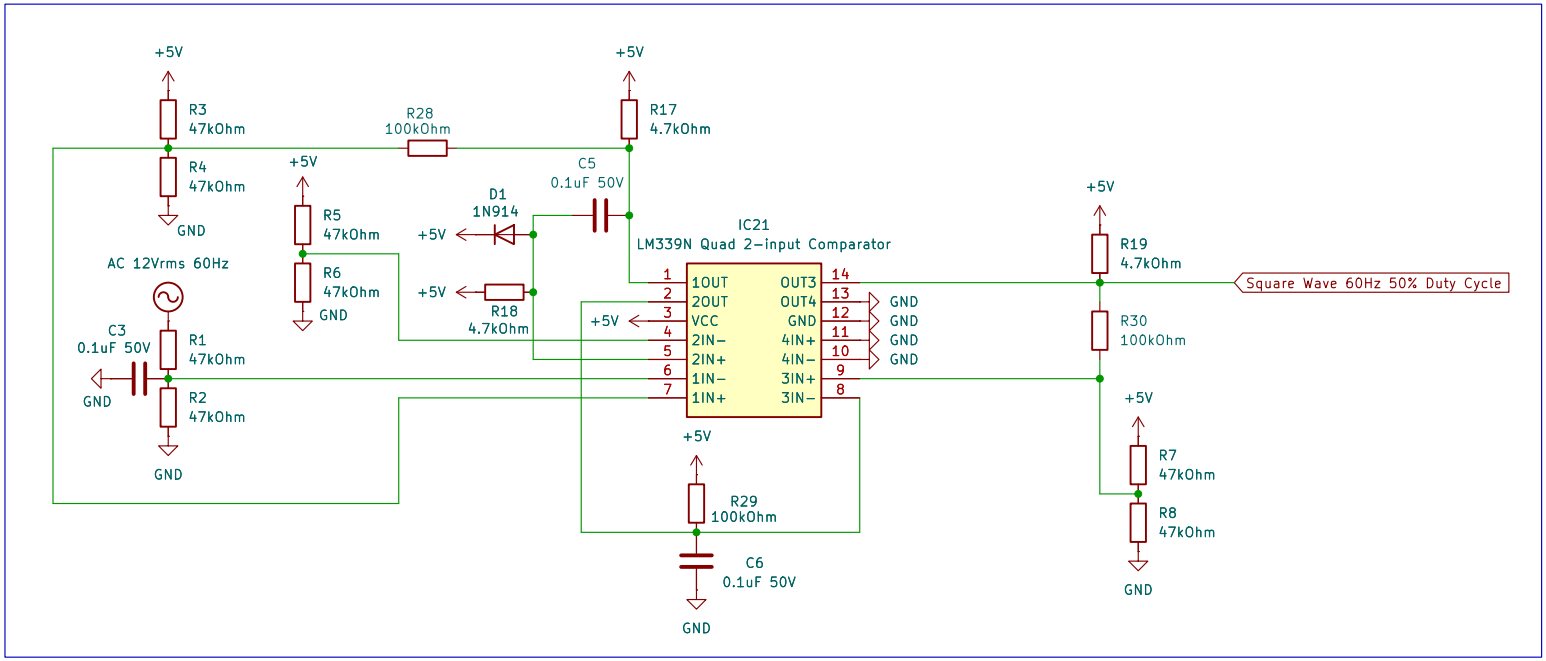
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /Power/
File: Power.kicad_sch

Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 2/9

Square Wave Generator



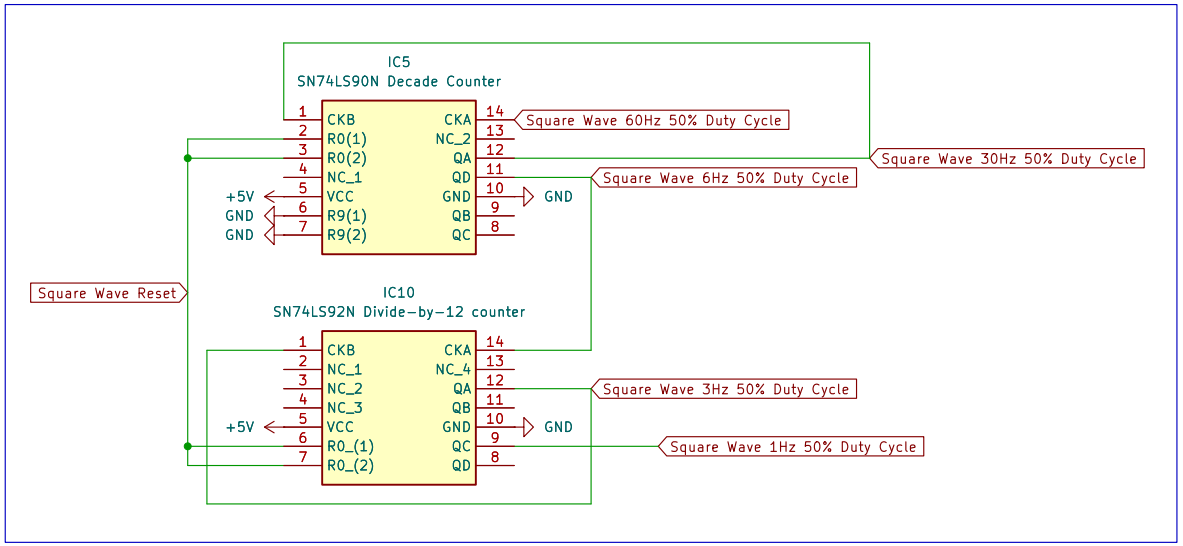
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /Square Wave Generator/
File: Square_Wave_Generator.kicad_sch

Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 3/9

1Hz Square Wave Generator



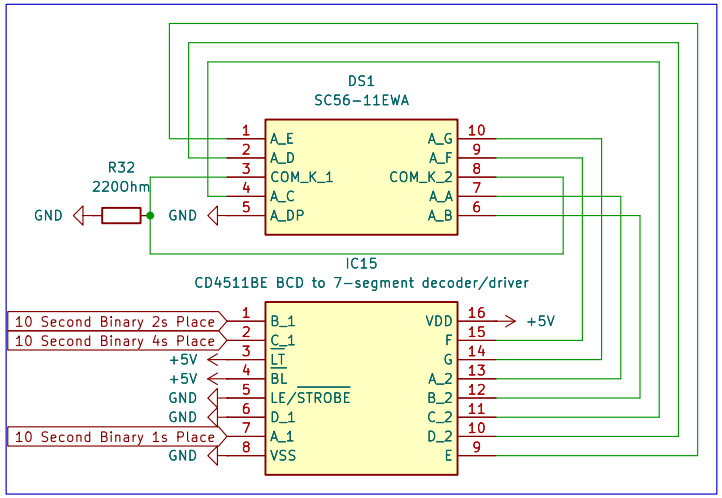
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /1Hz Square Wave Generator/
File: 1Hz_Square_Wave_Generator.kicad_sch

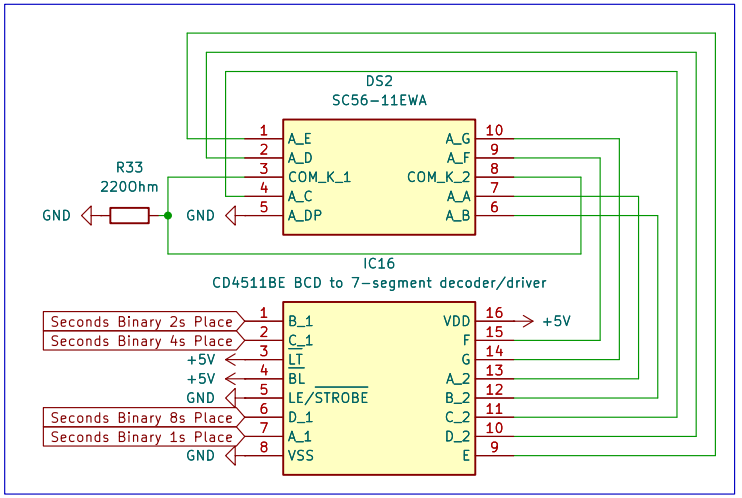
Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 4/9

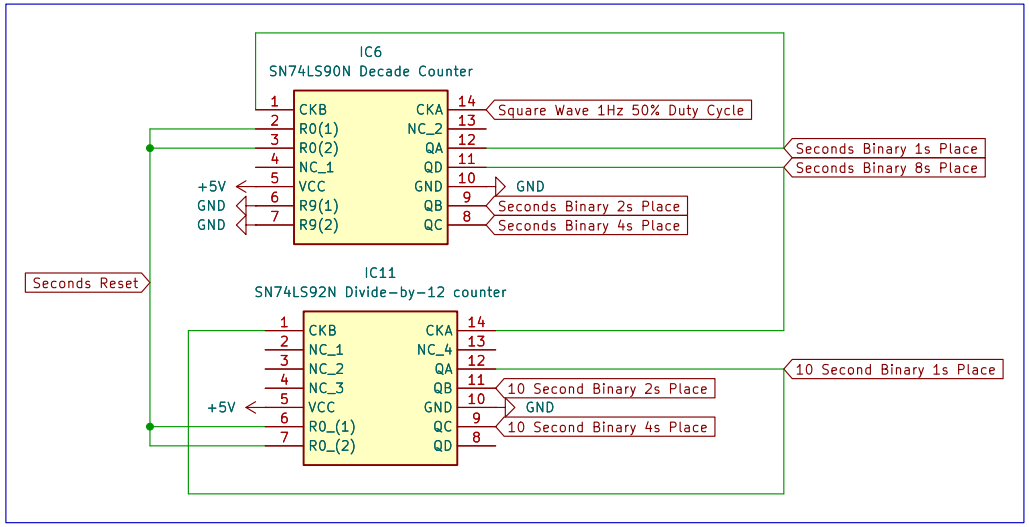
Seconds 10s Display



Seconds 1s Display



Seconds Counter



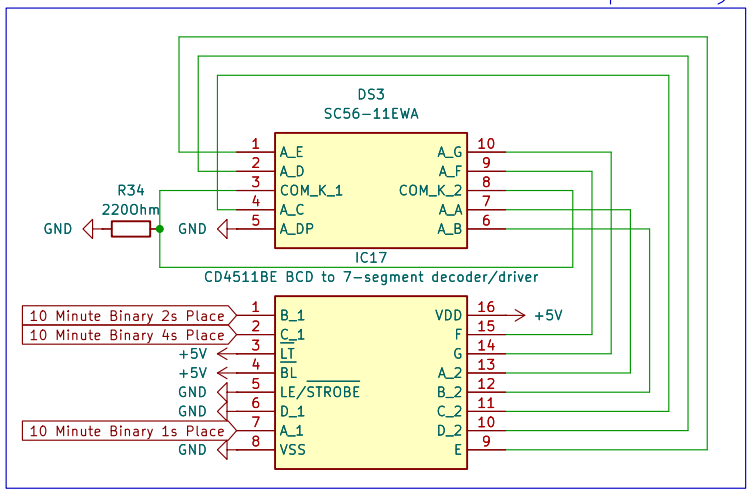
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /Seconds Counter/
File: Seconds_Counter.kicad_sch

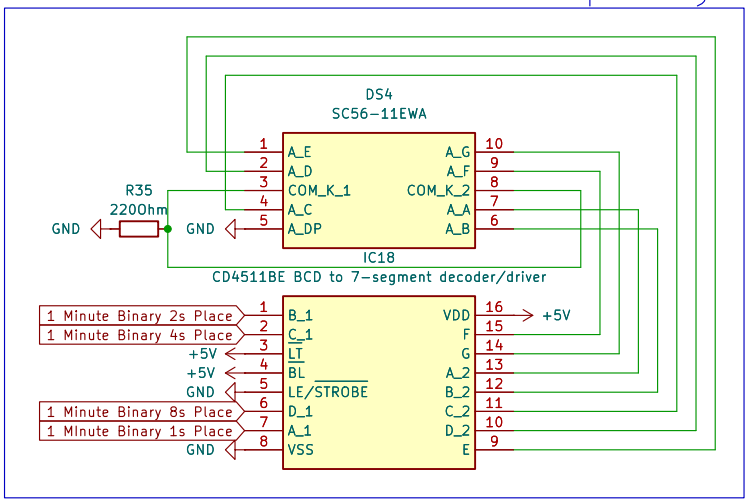
Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 5/9

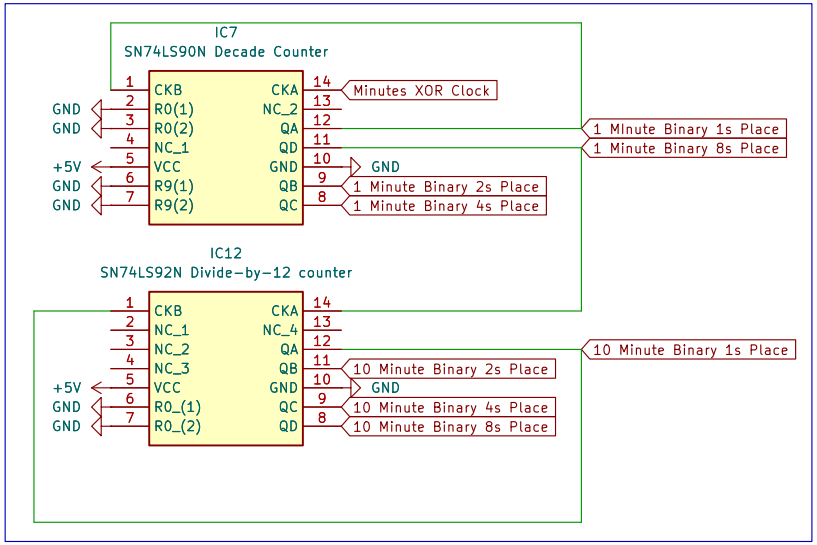
Minutes 10s Display



Minutes 1s Display



Minutes Counter



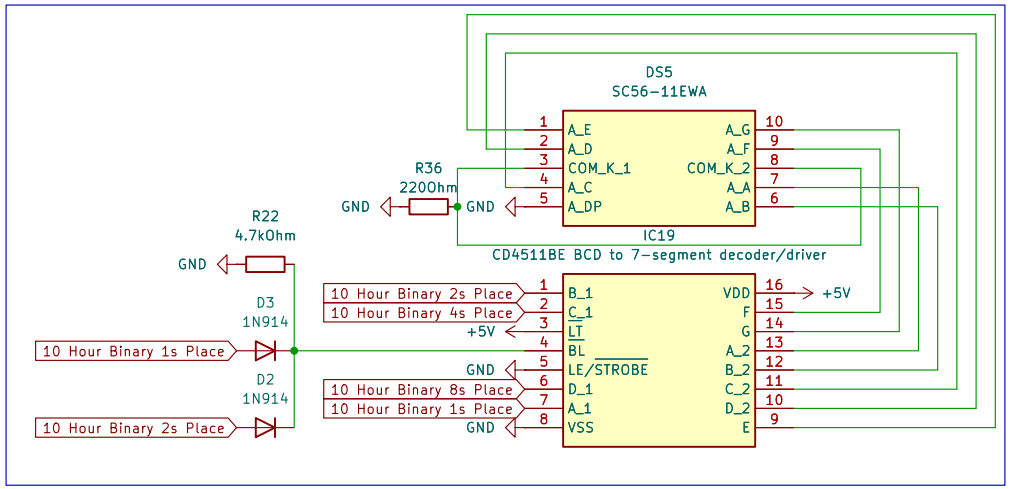
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /Minutes Counter/
File: Minutes_Counter.kicad_sch

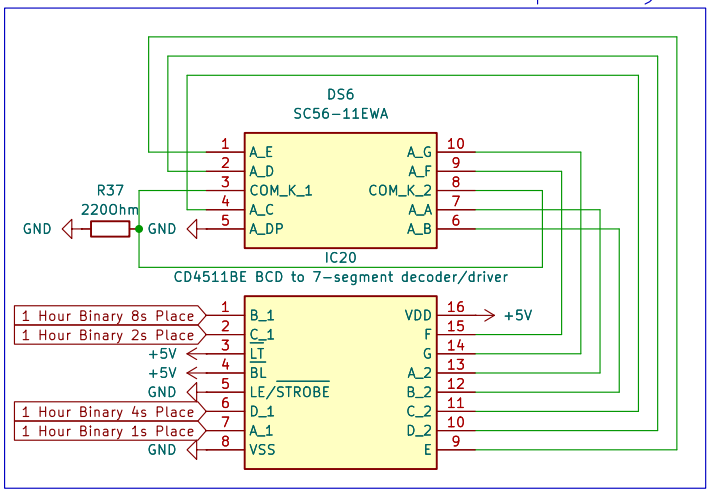
Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 6/9

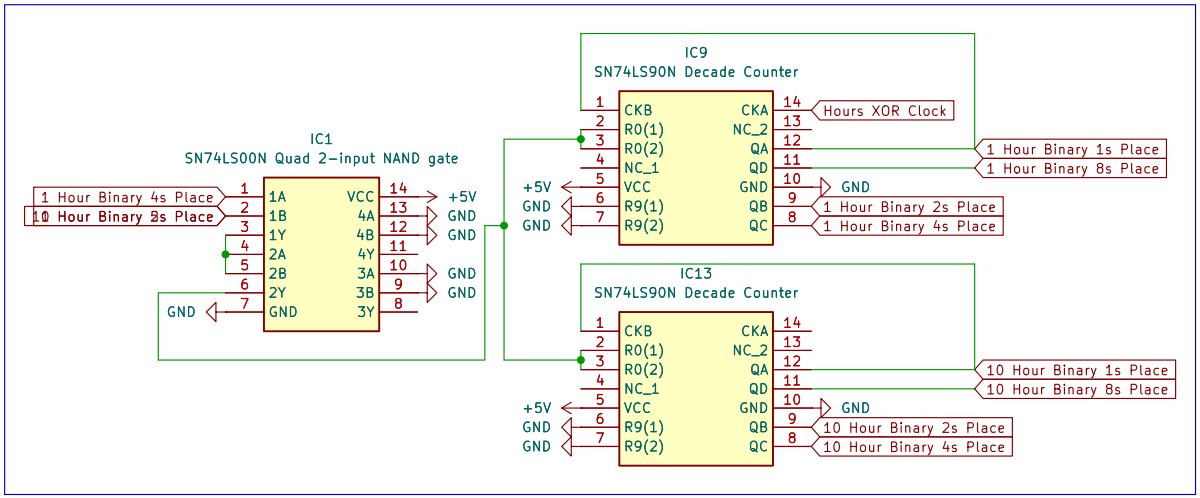
Hours 10s Display



Hours 1s Display



Hours Counter



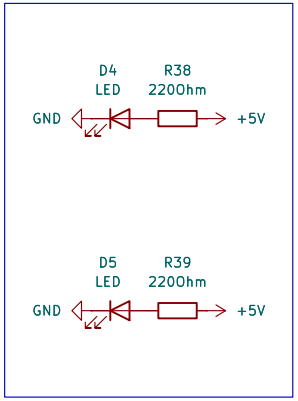
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /Hours Counter/
File: Hours_Counter.kicad_sch

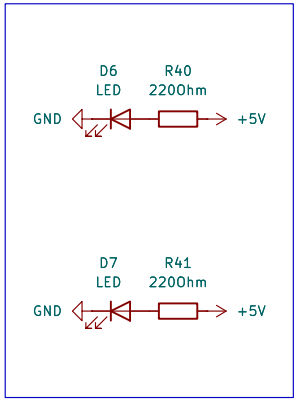
Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 7/9

Hours/ Minutes Colon



Minutes/ Seconds Colon



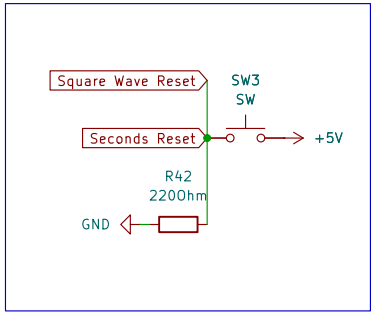
https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /Colon LEDs/
File: Colon LEDs.kicad_sch

Title: 24 Hour Breadboard Clock

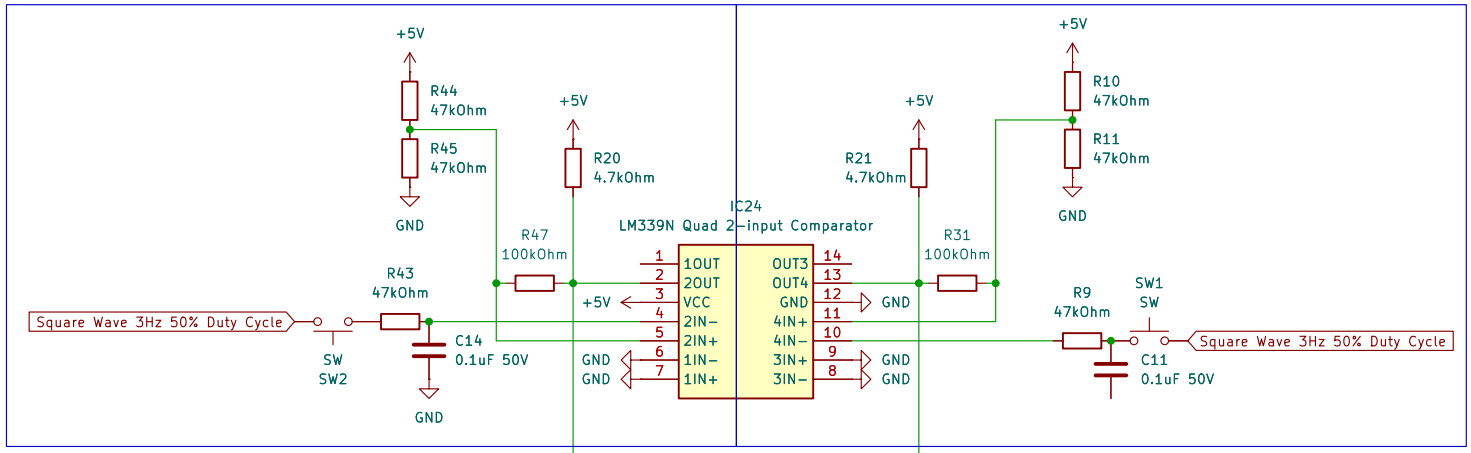
Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 8/9

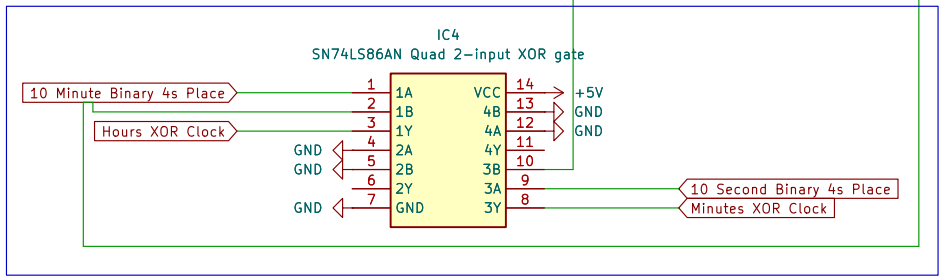
Seconds and Square Wave Reset



Minutes 3Hz Set Button

Hours 3Hz Set Button





Hour/Minute XOR Logic

https://www.youtube.com/watch?v=3XDH-fZKnQk
Clock design from Ben Eater's YouTube series
KiCAD schematics by Jost Strnat

Sheet: /Time Set Buttons/
File: Time_Set_Buttons.kicad_sch

Title: 24 Hour Breadboard Clock

Size: A4	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 9/9